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CA-R-05



GOVERNANCE & ETHICS

Watching People

A governance framework for workplace computer vision that survives scrutiny.

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October 2026 CloudyAI Research Free to download - no email required

SUMMARY

Machine vision inspects things; this inspects people. A practical framework covering purpose limitation, Zimbabwe's data protection Act, worker consultation, retention, and the red lines we will not cross.

There is a meaningful difference between a camera that checks whether a cap is on a bottle and a camera that checks whether a person is wearing a hard hat. The first inspects an object. The second inspects a human being at work, and the power relationship in that act changes what responsible deployment looks like.

This paper sets out the framework we use. It is written for plant managers and safety leads who want the safety benefit without turning the factory into a surveillance operation - and who understand that a system workers do not trust will be defeated by a piece of tape on a lens.

Six principles

1. Purpose limitation, written down

The system exists to reduce a specific safety risk. That purpose goes in writing, in plain language, before the first camera is configured - and anything outside it is out of scope. PPE compliance, yes. Individual productivity scoring, no. The purpose statement should fit on one page and be understandable to someone who has never heard of a neural network.

2. No biometric identification

Detecting a person is not identifying a person. We do not deploy facial recognition, gait recognition, or any mechanism that maps a detection to a named employee. Where a violation needs follow-up, the alert points to a zone, a time, and a clip that a supervisor reviews in the ordinary course of their work. This is slower than automatic attribution and considerably more defensible.

3. Minimum necessary data

The strongest privacy control is architectural: run inference at the edge and do not record video at all. Send events, counts, and short evidence clips. Where clips are needed, keep them for the shortest period that supports the safety process - typically 30 days - and restrict access to a named group. Raw continuous footage, if retained, needs a separate justification and a retention schedule.

4. Transparency

Signage at every entrance where cameras operate, a plain-language notice explaining what is detected and what is not, a named contact, and a briefing to workers and their representatives before go-live. Not a poster in the HR office - a conversation on the floor, in the languages people actually speak.

5. Human in the loop

The system produces observations. A person decides what they mean and what happens next. No automatic discipline, no automatic deduction, no automatic publication of individual results. Where a supervisor cannot explain an alert, the alert is treated as a system error to be investigated, not a worker error to be punished.

6. Reviewable and reversible

Document thresholds, model version, zones, and retention. Review the system with worker representatives every six months. Keep a documented way to switch it off - and mean it.

The Zimbabwean legal frame, briefly

The Cyber and Data Protection Act [Chapter 12:07] (2021) governs the processing of personal data in Zimbabwe, and images of identifiable people are personal data. The Act sets out conditions familiar from other data protection regimes: lawful and fair processing, purpose specification, minimality, accuracy, security safeguards, transparency, and respect for data subject rights. It also restricts cross-border transfers of personal data unless adequate protections are in place, and requires notification of the supervising authority in defined circumstances.

Practically, for a workplace vision deployment, that means: establish a lawful basis and document it; define and publish the purpose; collect only what the purpose requires; secure the data; tell people; have a process for access and complaints; and think carefully before sending footage to a cloud provider outside Zimbabwe. This is a summary for orientation, not legal advice - engage local counsel and, where required, the relevant authority.

> **A useful test**
>
> If a worker asked to see everything the system recorded about them, could you produce it, explain it, and defend why you kept it? If the answer is no, the design is not finished.

Before the pilot: the DPIA, in eight questions

01 What specific safety risk does this address, and what is the evidence that the risk is real?

02 What exactly is detected, and what is deliberately not detected?

03 Where does inference happen, and where does data go?

04 Who can see events and clips, and what is the access process?

05 How long is each category of data kept, and what deletes it?

06 How are workers informed, and who represents them in this decision?

07 What is the false-positive process, and what happens when the system is wrong?

08 How and when will this be reviewed, and how would it be stopped?

Vendor questions worth asking

Question	A good answer sounds like
Does the model run on site or in the cloud?	On site, with events forwarded. Video does not leave the plant.
Are you training on our footage?	Only with written agreement; here is the deletion process.
What is your false-positive rate on our site?	We measured it in shadow mode. Here is the distribution.
Which sub-processors touch the data?	A named list with locations, reviewed annually.
Can we export events and delete everything?	Yes. Here is the export format and the retention control.
How do you test for bias across uniforms, skin tone, and lighting?	A stratified validation set, with per-group recall reported.

Red lines

Some deployments are technically possible and simply not worth doing. We do not build: facial recognition or identity matching; emotion or fatigue detection from faces; individual productivity scoring from body movement; audio surveillance; covert monitoring; or anything that reports on workers outside defined safety zones and working hours.

This is partly ethical and partly practical. Systems in that territory attract disputes, regulator attention, and quiet sabotage, and they tend to be switched off within a year - after the money has been spent and the safety programme has lost credibility.

Making it last

The deployments that survive are boring. A clear purpose, a small scope, a visible benefit for the people being monitored, a supervisor who trusts the alerts, and a review meeting that actually happens. The technology is the easy part.

ABOUT THIS PAPER

CloudyAI is a Zimbabwean AI studio. We deploy computer vision on existing plant cameras, audit AI spend, and help teams respond when an AI system misbehaves. Research is published free, without an email wall.

Corrections and questions: research@cloudyai.co.zw. If you would like this method applied to your plant, the first conversation is free and the camera audit is a fixed-price week.